ASSIGNMENT 1

Identify the Association Between Dependent and Independent Variables

1. Aim

To identify and analyse the relationship between independent and dependent variables using correlation analysis and graphical visualisation.

2. Objectives

1)To understand independent and dependent variables.

2)To calculate the correlation coefficient between numerical variables.

3)To represent the relationship using a scatter plot.

4)To identify whether the relationship is positive, negative, or weak.

3. Theory

In data analysis and machine learning, variables are used to represent different characteristics of a dataset. Understanding the relationship between variables helps us identify patterns and make predictions.

An independent variable is an input or predictor variable used to study a relationship. A dependent variable is the outcome variable whose value is being studied.

For example, if we want to study the relationship between the number of hours a student studies and the marks obtained in an examination, study hours can be considered the independent variable, while marks are the dependent variable.

Correlation is a statistical measure used to describe the strength and direction of the linear relationship between two numerical variables. The Pearson correlation coefficient generally ranges from -1 to +1.

1)Positive correlation: Both variables tend to increase together.

2)Negative correlation: One variable tends to decrease as the other increases.

3)Zero correlation: No linear relationship is detected.

A scatter plot is a graphical representation in which each point represents one observation. It helps us visually understand the relationship between two variables.

Correlation does not necessarily prove causation because other factors may influence the variables.

4. Algorithm

1.Import the pandas and matplotlib libraries.

2.Create a sample dataset containing study hours and marks.

3.Display the dataset.

4.Calculate the correlation matrix.

5.Draw a scatter plot.

6.Observe and interpret the relationship between the variables.

5. Explanation of Important Code.

* pandas is used to store and analyse data in tabular form.

* matplotlib.pyplot is used to create graphs.

* pd.DataFrame() creates a dataset containing two columns.

* data.corr() calculates the correlation between numerical variables.

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6. Expected Output

The terminal displays the dataset and its correlation matrix. The scatter plot shows study hours on the X-axis and marks on the Y-axis.

The sample data demonstrate a positive relationship: marks generally increase as study hours increase.

7. Applications

1. Analysing the relationship between study hours and examination marks.
2. Studying the relationship between advertising expenditure and sales.
3. Analysing temperature and electricity consumption
4. Exploring relationships between product prices and demand.
5. Identifying patterns in business and scientific data.

8.Advantages

1.Correlation is easy to calculate and understand.

2.It helps identify relationships between v

3.It helps identify relationships between v

4.It can help identify useful variables for further analysis.

9. Limitations

1. Correlation does not prove causation.
2. Pearson correlation mainly measures linear relationships.
3. Outliers can affect the correlation coefficient.
4. Correlation alone cannot explain why a relationship exists.

10. Conclusion

The association between independent and dependent variables was analysed using correlation analysis and a scatter plot. The example demonstrated a positive relationship between study hours and marks. This experiment shows how statistical analysis can help identify patterns in data